PART 1 — ENGLISH · 第一部分 — 英文

Request for Budgetary Technical and Commercial Proposal No. OVE75-RFQ-25

Subject: Plate heat exchangers (outside the Basic Engineering scope).

KVANT LLC (Russian Federation), within the Basic Engineering (FEED) stage of the construction project of a copper production plant — a roasting–leaching–electrowinning complex with a capacity of 75,000 tonnes of copper cathodes per year — kindly requests your budgetary technical and commercial proposal for the item above. Project site: an operating metallurgical site in north-west Russia.

Item summary: Lean electrolyte heating 1,500 kW + electrolyte coolers 2 × 2,500 kW; sectional design with a standby section for cleaning of gypsum scale. Technical requirements are given in Annex 1 to this request.

Please include in your quotation:

1. budget price quoted separately on EXW (manufacturer’s works) and DDP project site, north-west Russia (Incoterms 2020), stating the currency and the validity period of the offer;
2. manufacturing lead time and estimated delivery time (weeks from the date of order / advance payment);
3. overall dimensions and weights of a unit and of the largest shipping blocks (for the logistics study);
4. reference list of similar supplies: product, process duty and parameters, year of supply, country;
5. materials of the parts in contact with the process medium, and the scope of supply (instrumentation, spare parts, documentation);
6. utility consumption and installation requirements (if available).

This is a budgetary enquiry at the Basic Engineering (FEED) stage: it does not constitute an offer and creates no obligations for either party; detailed technical data sheets will be issued at the next project stage.

Reply deadline: we kindly ask you to submit your quotation within 3 (three) weeks from receipt of this request to stepan@kvantpro.com. Please confirm receipt of this request.

Confidentiality. This request and the attached technical materials are provided solely for the preparation of the quotation, are confidential, and shall not be disclosed to third parties, copied or published without the prior written consent of KVANT LLC. Submission of a quotation creates no legal obligations for either party.

Contact person: Stepan Kharkov, KVANT LLC, stepan@kvantpro.com.

Annex 1: Technical requirements for the requested item.

Annex 1. Technical Requirements

To RFQ No. OVE75-RFQ-25 · The values follow the Customer's input data and terms of reference; Basic Engineering (FEED) stage; subject to refinement at the next project stage.

RFQ item	Plate heat exchangers (outside the BE scope)
Summary	Lean electrolyte heating 1,500 kW + electrolyte coolers 2 × 2,500 kW; sectional design with a standby section for cleaning of gypsum scale
Specific requirements	Provision for switching each section over to flushing with a special solution and with water; bypass lines for fine control

Equipment covered by this request:

1. **Lean electrolyte heater — heat exchange (outside the BE scope) · tag 4.1.17**

Duty	Heating of the recirculated solution by the leaching filtrate upstream of the cascade; start-up heating by steam
Key parameters	Design heat duty 1,240 kW; 50.0 → 59.4 °C at 124.7 m³/h; 75.0 → 65.6 °C at 129.5 m³/h
Quantity	1
Materials	Hastelloy G-35 or equivalent
Dimensions, hold-up	W = 1,500 kW; margin factor 1.2
Additional requirements	Plate type, sectional; standby section for section-by-section cleaning of gypsum scale; bypass lines for fine control

2. **Plate electrolyte cooler — electrolyte circulation · tag 4.2.2**

Type / model	Plate heat exchanger, water-cooled
Duty	Cooling of the rich solution before it is fed to the electrolysis cells
Key parameters	Design heat duty 3,772 kW; 50.0 → 48.2 °C at a process flow of 1,634.5 m³/h; 22.5 → 37.5 °C at a water flow of 216.1 m³/h
Quantity	2 (one per electrolysis series)
Materials	Steel 316L or equivalent
Dimensions, hold-up	W = 2,500 kW per train; actual 5,000 kW; margin factor 1.3
Additional requirements	The heat exchangers are not included in the Basic Engineering scope of Lot No. 3 (the "outside BE" list); their parameters are set by the input data

The quotation must confirm: process medium, temperature, pressure, materials of wetted parts, capacity, dimensions and weight, power consumption, manufacturing lead time, delivery terms.

PART 2 — CHINESE · 第二部分 — 中文

预算性技术商务报价邀请函 编号 OVE75-RFQ-25

事由:板式换热器(基础设计范围外)。

KVANT LLC(俄罗斯联邦)正在开展一座铜冶炼厂建设项目的基础设计(FEED)——"焙烧—浸出—电积"联合工艺装置,阴极铜产能为每年 75,000 吨。现请贵方就上述设备提交预算性技术商务报价。项目场地:俄罗斯西北部一座在运营的冶金厂区。

设备简要特性:贫电解液加热 1,500 kW + 电解液冷却器 2 × 2,500 kW;分段式结构,设有用于清除石膏结垢的备用段。 技术要求见本函附件 1。

请在报价中包含:

1. 预算价格——分别按 EXW(制造商仓库)及 DDP 项目场地(俄罗斯西北部)条件(Incoterms 2020)报价,并注明币种及报价有效期;
2. 制造周期及预计交货期(自订单/预付款之日起,按周计);
3. 单台设备及最大运输单元的外形尺寸和重量(用于物流研究);
4. 类似供货业绩表:产品、工艺介质及参数、供货年份、国家;
5. 与工艺介质接触部件的材质,以及供货范围(仪表、备件、技术文件);
6. 能耗及安装要求(如有资料)。

本询价为基础设计(FEED)阶段的预算性询价:不构成要约,对双方均不产生义务;详细技术数据表将在项目下一阶段发出。

回复期限:请于收到本函后 3(三)周内将报价发送至 stepan@kvantpro.com。请确认收到本函。

保密条款。本函及随附技术资料仅供编制报价使用,属保密信息,未经 KVANT LLC 事先书面同意,不得向第三方披露、复制或发布。提交报价不构成双方的法律义务。

联系人：Stepan Kharkov, KVANT LLC, stepan@kvantpro.com。

附件 1:本询价设备的技术要求。

(职务) (姓名) (签字)

附件 1. 技术要求

对应询价函编号 OVE75-RFQ-25 · 参数依据业主原始数据及设计任务书;基础设计(FEED)阶段,后续阶段可能进一步细化。

询价设备	板式换热器(基础设计范围外)
简要特性	贫电解液加热 1,500 kW + 电解液冷却器 2 × 2,500 kW;分段式结构,设有用于清除石膏结垢的备用段
特殊要求	每段应可切换至用专用溶液和水进行清洗;设旁路管线以实现精细调节

本询价涵盖的设备清单:

1. **贫电解液加热换热器——换热(基础设计范围外) · 位号 4.1.17**

用途	在梯级之前用浸出滤液加热循环溶液;开车时用蒸汽加热
主要参数	计算热负荷 1,240 kW;50.0 → 59.4 °C,流量 124.7 m³/h;75.0 → 65.6 °C,流量 129.5 m³/h
数量	1

材质	Hastelloy G-35 或同等材质
尺寸、持液量	W = 1,500 kW;裕度系数 1.2
附加要求	板式分段结构;设备用段,可逐段清除石膏结垢;设旁路管线以实现精细调节

2. **板式电解液冷却器——电解液循环 · 位号 4.2.2**

型号	板式换热器,水冷
用途	富液送入电解槽前的冷却
主要参数	计算热负荷 3,772 kW;工艺流量 1,634.5 m³/h,50.0 → 48.2 °C;冷却水流量 216.1 m³/h,22.5 → 37.5 °C
数量	2(每个电解系列 1 台)
材质	316L 钢或同等材质
尺寸、持液量	每条线 W = 2,500 kW;实际 5,000 kW;裕度系数 1.3
附加要求	3 号标段基础设计范围不包含换热器(列入"基础设计范围外"清单),但其参数已由原始数据确定

报价中必须确认:工艺介质、温度、压力、接触部件材质、生产能力、尺寸与重量、能耗、制造周期、交货条件。